ISTANBUL UNIVERSITY-CERRAHPAŞA
Department of Computer Engineering
BIMU2059-DIFFERENTIAL EQUATIONS
Midterm – Fall 2023

Date of Exam: 28.11.2023 Name:
Time: 15:20-16:20 Student ID:

1) Find the general and singular solution of the following differential equation:

$$y' = \frac{-xy^3}{(y+1)e^{-x}} \ .$$

(25 Pts)

2) Find the general solution to the following differential equation:

$$y' = -\frac{y^2 + xy^3}{y^3 siny + 5y^2 - xy}$$

(25 Pts)

3) Solve the following initial-value problem (IVP):

$$(1 + y')cos(x + y) = yy', \qquad y(0) = 1.$$

(25 Pts)

4) a) In which cases is the Bernoulli differential equation

$$y' + p(x)y = q(x)y^n$$

a linear differential equation?

b) Classify the given differential equations below (linear, exact,..., etc.).

i) $(x^2 + y^2 + x^2y^2 + 1)dx = (ycos^2x + y)dy$

ii) $y' = \frac{(x+y)^2}{1-2xy-x^2}$

iii) $((y + 1)x^2 - x^3y^2 - 8)dx + x^2dy = 0$

(25 Pts)

Dr. Uğur Odabaşı